

**Time:** 5 minutes**Outcome:** Analyze Short-Run Cost Curves: Relationships & Shifts.**Difficulty:** ●●○ Intermediate

Short-Run Cost Curves: Relationships & Shifts



THE CORE IDEA

Short-run cost curves move differently because fixed costs affect AFC and ATC, while variable-input costs affect AVC, ATC, and MC. A below-average marginal observation pulls the average down. Labor is the variable input in the standard short-run model. A fixed-cost increase raises fixed cost per unit and total cost per unit but leaves variable and marginal cost unchanged.



HOW TO RECOGNIZE IT

- A fixed-cost increase raises fixed cost per unit and total cost per unit but leaves variable and marginal cost unchanged.
- A technology improvement reduces the variable input needed at every output.
- Labor is the variable input in the standard short-run model.
- A firm receives a property-tax increase that does not depend on output.



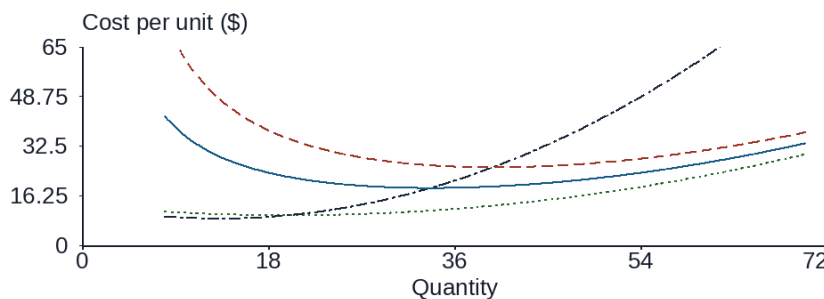
WATCH OUT

A common mistake is to mix fixed and variable shifts. A fixed-cost increase raises fixed cost per unit and total cost per unit but leaves variable and marginal cost unchanged.



WORKED EXAMPLE: SHIFTING SHORT-RUN COST CURVES

... AVC unchanged -- MC unchanged — ATC: original - - - ATC: higher fixed cost



Suppose the firm signs a more expensive lease while output and variable input prices are unchanged. The higher fixed cost raises average fixed cost and therefore shifts ATC upward. AVC and MC remain in the positions shown because neither depends on fixed cost. The production decision at the margin is unchanged.



CHECK YOURSELF

With fixed cost unchanged, why does the ATC-AVC gap shrink as output rises? How is that different from an increase in the lease payment at a fixed output?

**READY?**

Return to the game and master this concept.